Xingyu Zhao

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Education

University of Wisconsin-Madison Sep 2024 - Present
B.S. in Mathematics and Computer Sciences (Honors)
Expected Dec 2026 | GPA: **4.00/4.00**

Fudan University Sep 2022 - Jun 2024
Undergraduate studies in Computer Science
GPA: **3.90/4.00** | Class rank: **1/88**

Research Interests

Statistical machine learning, optimization, and learning dynamics; generalization and reasoning in large language models (LLMs); LLM interpretability and alignment; multi-agent interaction and behavioral transfer.

Publications and Manuscripts

[1] **Shattered Compositionality: Counterintuitive Learning Dynamics of Transformers for Arithmetic.**
Xingyu Zhao, Darsh Sharma, Rheeya Uppaal, and Yiqiao Zhong.
NeurIPS 2026, accepted arXiv:2601.22510 | Code

[2] **Cross-Trait Transfer in Subliminal Learning.**
Xingyu Zhao and Yiqiao Zhong.
Under review at ICLR 2027 arXiv:2610.04260 | Code

Research Experience

University of Wisconsin-Madison 2025 - Present
Undergraduate Researcher, Department of Statistics | Advisor: Prof. Yiqiao Zhong

Transformer Learning Dynamics and Compositional Generalization

- Tracked subskill acquisition in synthetic addition, multiplication, comparison, and sorting tasks; identified reverse-order and parallel learning patterns and examined their implications for compositional generalization.
- Developed diagnostic metrics based on conditional mutual information to explain reverse-order learning; designed targeted ablation studies to test how training-data correlations support later-learned subskills.
- Trained NanoGPT transformers from scratch at approximately 10M, 20M, and 100M parameters and fine-tuned Pythia-1B; studied positional encodings, output formats, and scratchpad supervision.
- Characterized mixing errors under distribution shift and evaluated reasoning robustness in 10 pretrained LLMs using 50 clause-extended GSM8K templates.

Research Experience *(continued)*

University of Wisconsin-Madison | Advisor: Prof. Yiqiao Zhong

Cross-Trait Transfer and Mechanisms of Subliminal Learning

- Studied how behavioral traits transfer from a teacher language model to a student model through training on semantically unrelated outputs, including numbers, symbols, mathematical solutions, and code.
- Developed external metrics of output-probability shifts and internal metrics based on activation steering and representation geometry; compared them with behavioral transfer matrices to characterize statistical patterns of trait transmission.
- Fine-tuned 7-8B instruction-tuned models across the Qwen, Llama, Mistral, and OLMo families using LoRA; separately compared supervised fine-tuning and distribution distillation.
- Designed multi-trait and rare-trait experiments to study feature competition among behavioral traits.
- Found that related traits can be amplified more than the teacher's own trait; used increasing effective rank of transfer matrices to characterize the shift from shared behavioral changes to more differentiated preferences.

Selected Coursework

Mathematics: Real Analysis; Advanced Linear Algebra; Abstract Algebra; Stochastic Processes; Stochastic Calculus; Optimization (introductory course in linear, integer, and mixed-integer programming).

Theoretical computer science: Algorithms; Data Structures; Theory of Computation.

Machine learning: Deep Learning; Reinforcement Learning.

Systems and graphics: Parallel Programming; Computer Systems; Database Systems; Computer Graphics.

Research Methods and Technical Skills

Statistical methods: Information-theoretic and distributional analysis; Fisher geometry; PCA/SVD; permutation tests with Holm correction for multiple comparisons.

Experimental methods: Controlled experiments and ablation studies; latent activation analysis and steering; learning-dynamics and out-of-distribution evaluation.

Scientific computing: Python; PyTorch; Hugging Face Transformers; Unsloth; NumPy; pandas; Matplotlib.

Programming and tools: C/C++; Java; JavaScript; SQL; Bash; Git; Linux; pytest; LaTeX.

Selected Software Projects

B-Spline Interactive Demo GitHub

Computer Graphics course project | JavaScript

- Implemented an interactive B-spline visualizer using Cox-de Boor recursion, draggable control points, configurable degree and knot vectors, and real-time piecewise-basis visualization.
- Adopted as an interactive learning tool and lecture aid in Prof. Michael Gleicher's CS559 Computer Graphics course.

Fudan University Study Room Booking Prototype GitHub

Django | SQLite | Bootstrap

- Built a full-stack study-room reservation prototype with server-side booking workflows, persistent relational storage, and a responsive user interface.